AQA 3.1.12 Acids and bases

AQA A-level Chemistry · Revision questions · Sheet 5 of 8

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- $pH = -\log_{10}[H^+]$ and $[H^+] = 10^{-pH}$
- $K_w = [H^+][OH^-]$ (1.00×10^{-14} mol^2 dm^{-6} at 25 °C); for a base $[H^+] = K_w \div [OH^-]$
- Weak acid on its own: $[H^+] = \sqrt{(K_a \times [HA])}$; $K_a = 10^{-pKa}$
- Buffer: $[H^+] = K_a \times [HA] \div [A^-]$ — use moles, the volume cancels
- Weak acid + NaOH: **K_a = [H$^+$] × salt ÷ acid**, where salt = mol NaOH added and acid = mol HA left

Section A: Short-answer questions

01 Calculate the pH of a 0.0250 mol dm^{-3} solution of barium hydroxide at 25 °C. Give your answer to 2 decimal places. $K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6} at 25 °C

[2 marks]

02 State the meaning of the term *strong acid*.

[1 mark]

03 A buffer contains ethanoic acid and sodium ethanoate. Explain how this buffer resists a change in pH when a small amount of acid is added.

[2 marks]

04 What is the pH of a 0.46 mol dm^{-3} solution of potassium hydroxide at 298 K? ($K_w = 1.0 \times 10^{-14}$ mol^2 dm^{-6} at 298 K)

[1 mark]

A 0.34
B 13.66
C 13.96
D 14.34

05 Calculate the concentration of hydrogen ions in a solution with pH = 3.40

[1 mark]

06 A buffer solution contains equal concentrations of ethanoic acid and sodium ethanoate. Calculate the pH of this buffer solution. For ethanoic acid, $K_a = 1.74 \times 10^{-5}$ mol dm^{-3}

[2 marks]

Section B: Longer questions

07 Ethanoic acid is a weak acid.

07.1 Give the expression for the acid dissociation constant (K_a) for ethanoic acid (CH_3COOH).

[1 mark]

07.2 A buffer solution contains 0.025 mol of sodium ethanoate dissolved in 500 cm^3 of 0.0700 mol dm^{-3} ethanoic acid at 25 °C. A sample of 5.00 cm^3 of 2.00 mol dm^{-3} hydrochloric acid is added to this buffer solution.

Calculate the pH of the solution formed. For ethanoic acid, $K_a = 1.76 \times 10^{-5}$ mol dm^{-3} at 25 °C

[5 marks]

08 36.25 cm^3 of 0.200 mol dm^{-3} sodium hydroxide solution are added to 25.00 cm^3 of 0.150 mol dm^{-3} hydrochloric acid.

Calculate the pH of the final solution at 25 °C. $K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6} at 25 °C

[5 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	M1 $[OH^-] = 2 \times 0.0250 = 0.0500$ mol dm^{-3} M2 pH = 12.70	*Source: Practice question (written for this set)* pH = 12.40 (not × 2) scores 1	2
02	Completely dissociates / ionises (to form H^+ ions)	*Source: 7405/1 Jun 2018 Q05.1*	1
03	M1 $CH_3COO^- + H^+ \rightarrow CH_3COOH$ / added H^+ reacts with ethanoate ions (large reservoir from the salt) M2 ratio $[CH_3COOH] \div [CH_3COO^-]$ stays almost constant, so $[H^+]$ stays almost constant	*Source: Practice question (written for this set)*	2
04	**B**	*Source: 7405/3 Jun 2017 Q10*	1
05	$[H^+] = 10^{-3.40} = 3.98 \times 10^{-4}$ mol dm^{-3}	*Source: Practice question (written for this set)*	1
06	M1 $[H^+] = K_a$ (as $[HA] = [A^-]$) M2 pH = 4.76	*Source: Practice question (written for this set)*	2
07.1	$K_a = [CH_3COO^-][H^+] \div [CH_3COOH]$	*Source: 7405/1 Jun 2018 Q05.5–05.6*	1
07.2	M1 n(HCl) added = 0.01 mol **and** $n(CH_3COOH) = 0.035$ mol M2 $n(CH_3COO^-) = 0.025 - 0.01$ (= 0.015) M3 $n(CH_3COOH) = 0.035 + 0.01$ (= 0.045) M4 $[H^+] = 1.76 \times 10^{-5} \times 0.045 \div 0.015$ ($= 5.28 \times 10^{-5}$) M5 pH = 4.28	If 0.07 mol used for CH_3COOH can score M2, M3, M4 (pH = 4.03) M4 conditional on an attempt at addition / subtraction in M2 or M3 M5 must be 2 dp	5
08	M1 $n(OH^-) = 7.25 \times 10^{-3}$ mol **and** $n(H^+) = 3.75 \times 10^{-3}$ mol M2 excess $OH^- = 3.50 \times 10^{-3}$ mol M3 $[OH^-] = 3.50 \times 10^{-3} \div 61.25 \times 10^{-3}$ ($= 5.71 \times 10^{-2}$) M4 $[H^+] = 1.00 \times 10^{-14} \div 5.71 \times 10^{-2} = 1.75 \times 10^{-13}$ M5 pH = 12.76	*Source: 7405/1 Nov 2021 Q06.8* Allow pH = 12.8 Alternative: pOH = 1.24, pH = 14 – 1.24	5